

Tanmay Gupta

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EDUCATION

National Institute of Technology, Kurukshetra
Bachelor of Technology in Artificial Intelligence and Machine Learning, GPA: 8.21
July 2024 to May 2028

Haryana, India

SKILLS SUMMARY

Languages: Python, C++, JavaScript, SQL, Rust (Basic Proficiency)

Frameworks / Libraries: Node.js, Express.js, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Rasterio, GeoPandas, Sequelize, Hugging Face Transformers

Databases: MySQL, PostgreSQL, SQLite, Redis

Tools: Git, Docker, RabbitMQ, gRPC, Protocol Buffers, Ollama

CS Fundamentals: Operating Systems, Computer Networks, DBMS, System Design

PROJECTS

SynapseAI: Autonomous Multi-Agent Framework

May 2026 to Jun 2026

GitHub • Python • Ollama • Docker • Pydantic

- Architected an autonomous platform utilizing 7 specialized AI agents to automate EDA, feature engineering, and model selection, reducing the manual ML benchmarking lifecycle from days to minutes.
- Designed a centralized Blackboard Architecture using Pydantic for thread-safe state management, deterministic checkpointing, and structured inter-agent communication.
- Developed an orchestration supervisor agent to manage multi-stage agent workflows featuring dynamic conditional routing, automated retry logic, and fallback error recovery.
- Automated benchmarking across 9 algorithm families (Logistic Regression, Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost, SVM, KNN, MLP), with automatic classification/regression detection, metric selection, and leaderboard ranking.

Flyh: Distributed Flight Booking System

Jun 2025 to Dec 2025

GitHub • Node.js • Express.js • MySQL • RabbitMQ • Docker

- Developed a distributed airline reservation platform using 4 microservices, enabling modular deployment and fault isolation.
- Mitigated concurrent overbooking risks by implementing ACID-compliant MySQL transactions and explicit row-level locking via Sequelize, ensuring strict data consistency under high-concurrency booking simulations.
- Built an event-driven notification service using RabbitMQ for booking confirmations, cancellations, and reminders.
- Optimized ingress traffic by building a custom API Gateway featuring JWT-based stateless authentication, centralized request routing, and rate limiting to prevent downstream service degradation.

Landslide Prediction and Risk Mapping System

Jan 2026 to Mar 2026

GitHub • Python • TensorFlow • XGBoost • Rasterio

- Optimized geospatial data pipelines for a 163M+ pixel DEM grid, reducing terrain feature extraction time by 95.8% (from 2 hours to 5 minutes) by replacing iterative Python loops with vectorized NumPy gradient operations and 3x3 moving-window filters (via scipy.ndimage) running in compiled C code.
- Constructed a dataset of 56K+ samples with 18 engineered features using DEM, OSM, NDVI, and rainfall data.
- Engineered spatial proximity features using R-tree indexing (Shapely STRtree) to compute nearest-neighbor distances to roads, buildings, and rivers across 56K+ points, avoiding brute-force geometry comparisons.
- Trained Random Forest, XGBoost, and MLP models, achieving 67.9% accuracy for multi-class landslide prediction.

Geo-Distributed Search Engine

Mar 2026 to May 2026

GitHub • Python • gRPC • Protocol Buffers

- Designed a geo-distributed search engine with dynamic replica provisioning to reduce regional search latency.
- Implemented a fault-tolerant architecture with Master, Backup, Crawler, and Replica nodes supporting automatic failover via heartbeat checks and two-phase commit for metadata consistency.
- Developed idle-time-based replica eviction and adaptive replication to optimize storage utilization and query performance.
- Built distributed communication using gRPC and Protocol Buffers for metadata synchronization and request routing.

POSITIONS OF RESPONSIBILITY

Open Code Developers, NIT Kurukshetra

Aug 2024 to Present

Member

- Conducted workshops on Binary Search and Digit DP, equipping members with techniques frequently used in competitive programming contests.
- Designed a comprehensive competitive programming roadmap integrating CSES, USACO Guide, CP Algorithms, and Codeforces into a single learning path from beginner to ICPC-level preparation.

ACHIEVEMENTS

- Solved 1300+ algorithmic problems across major competitive programming platforms.
- LeetCode: Guardian (Max Rating: 2150), Best Global Ranks: 293, 537, 747 in Weekly Contests.
- Codeforces: Specialist (Max Rating: 1504), Best Rank: 732 in Educational Round 190.
- CodeChef: 3-star (Max Rating: 1663).